Beyond Stance Detection: Can Cause–Solution Annotation for Bengali Climate News Transfer Zero-Shot to Hindi?

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Abstract

Climate change reporting in South Asian languages remains dominated by stance and sentiment framing, leaving the underlying *causes* of environmental degradation and their corresponding *policy solutions* largely unannotated. We extend an existing Bengali climate news corpus with a 16-category cause taxonomy designed around Bangladesh's specific environmental exposure – spanning vehicular and industrial emissions, waste mismanagement, and climate-driven categories – rather than adapting a generic global taxonomy, and pair each cause with concrete, source-grounded solutions. To the best of our knowledge, this constitutes the first cause-and-solution annotation framework for Bengali climate news. Causes are scored via GPT-4o zero-shot annotation and validated against 700 human-annotated samples, achieving a Cohen's κ of 0.85. We benchmark five transformer backbones on the resulting multi-label classification task, with IndicBERTv2 achieving the strongest performance (F1-macro 0.730). Motivated by the absence of a comparable Hindi resource, we construct and release a new 2,000-article Hindi climate news dataset annotated with cause labels across the same 16-category taxonomy – to the best of our knowledge, the first Hindi dataset of its kind – and use it to test zero-shot cross-lingual transfer of our best Bangla-trained classifier. We find a substantial performance collapse under transfer (F1-macro 0.730 $\rightarrow$ 0.396), showing that strong in-language multi-label cause classification does not guarantee cross-lingual generalization, even between related, jointly-pretrained Indic languages. We release both datasets, the cause-solution framework, and all model checkpoints.

1 Introduction

Bangladesh is repeatedly ranked among the countries most affected by climate-related extreme weather (Eckstein et al., 2021). Existing Bengali climate NLP, however, targets stance and multi-perspective detection (Wasi et al., 2024) rather than the *causes* of environmental degradation or the *solutions* available to address them, and no comparable resource exists for Hindi. A recent Hindi retrieval benchmark includes a translated climate subset (Acharya et al., 2025), but targets claim-verification retrieval over machine-translated text rather than cause attribution over original Hindi journalism.

We introduce a 16-category, Bangladesh-specific cause taxonomy extending the Dhoroni corpus, pairing each cause with a source-grounded solution – to our knowledge the first such framework for Bengali. Causes are scored via GPT-4o zero-shot annotation (OpenAI, 2024) and validated against 700 human annotations (Cohen's $\kappa = 0.85$; Cohen, 1960). We benchmark five transformer backbones on the resulting multi-label task, with IndicBERTv2 (Doddapaneni et al., 2023) performing best (F1-macro 0.730). We further construct a 2,000-article Hindi climate news dataset annotated under the same taxonomy – to our knowledge the first of its kind – and test zero-shot cross-lingual transfer of our best Bangla-trained classifier. Despite both languages sharing a multilingual encoder, transfer largely fails: F1-macro drops from 0.730 in-language to 0.396 zero-shot, indicating that multilingual pretraining alone does not guarantee generalization on structured, culturally grounded classification (Pires et al., 2019; Conneau et al., 2020).

Our contributions:

1. A 16-category, Bangladesh-specific cause taxonomy extending Dhoroni, annotated via GPT-4o and validated against 700 human annotations ($\kappa = 0.85$) – to our knowledge, the first cause-and-solution framework for Bengali climate discourse.

2. A source-grounded solution layer paired with

each cause category.

3. A five-backbone benchmark on multi-label cause classification; IndicBERTv2 performs best (F1-macro 0.730).

4. A new Hindi climate news dataset (2,000 articles) annotated under the same 16-category taxonomy – to our knowledge, the first Hindi resource of its kind.

5. A zero-shot cross-lingual transfer evaluation showing a substantial performance drop (0.730 $\rightarrow$ 0.396), highlighting the limits of cross-lingual generalization for structured, culturally grounded classification.

2 Related Work

Climate NLP. Most NLP work on climate text targets domain-adaptive pretraining and claim verification rather than cause or solution extraction. ClimateBERT (Webersinke et al., 2022) continues pretraining on climate corpora for downstream classification and sentiment tasks, while CLIMATE-FEVER (Diggelmann et al., 2020) focuses on verifying real-world climate claims against Wikipedia evidence. For Bengali specifically, Dhoroni (Wasi et al., 2024) annotates stance and multi-perspective framing in climate news but does not label causes or solutions. We build directly on Dhoroni's corpus while introducing a task – structured, multi-label cause-solution classification – that none of these resources support.

Cross-lingual transfer for Indic languages. Multilingual encoders such as mBERT (Devlin et al., 2019) exhibit meaningful, if uneven, zero-shot cross-lingual transfer (Pires et al., 2019), and scaled pretraining (XLM-R; Conneau et al., 2020) improves this further. India-specific encoders, including MuRIL (Khanuja et al., 2021) and IndicBERTv2 (Doddapaneni et al., 2023), were built to close this gap for Indic languages, and IndicBERTv2 is evaluated on the general-purpose IndicXTREME suite. A separate line of work benchmarks Hindi retrieval, including a machine-translated climate subset (Acharya et al., 2025), but does not address cause attribution over original Hindi text. We instead evaluate cross-lingual transfer directly on a structured, culturally grounded classification task, where – unlike on general NLU benchmarks – we find transfer largely fails.

LLMs as annotators. Large language models are increasingly used to replace or supplement human annotation, often matching or exceeding crowd-worker accuracy on stance, topic, and frame-detection tasks (Gilardi et al., 2023). We adopt this approach for scalability, using GPT-4o (OpenAI, 2024) for zero-shot cause scoring in both languages, but depart from prior work by explicitly validating LLM labels against human annotation (Cohen's $\kappa = 0.85$; Cohen, 1960) rather than treating LLM output as ground truth.

3 Methodology

3.1 Cause Taxonomy and Solution Framework

We define a 16-category cause taxonomy for Bengali climate news, extending the Dhoroni corpus (Wasi et al., 2024) with cause- and solution-level annotation absent from the original release. Categories were designed manually rather than adapted from an existing global taxonomy, prioritizing causes with direct relevance to Bangladesh's environmental exposure profile: vehicular and industrial emissions, brick-kiln pollution, open waste burning, construction dust, industrial and sewage waste discharge into waterways, plastic waste (including single-use plastics), inadequate waste management, deforestation, and climate-driven categories such as sea-level rise, extreme weather events, and land-use change.

Each cause category is paired with a corresponding solution, grounded in authoritative sources rather than generated freely: guidelines from Bangladesh's Department of Environment (DoE), IPCC mitigation and adaptation recommendations, and the Bangladesh Delta Plan 2100 (BDP2100). We prioritize source traceability over solution novelty – each solution entry is tagged with its originating source type (DoE / IPCC / BDP2100 / local government guidance) so that claims can be independently verified rather than treated as model-generated suggestions. Solutions were manually paraphrased directly from these source documents, rather than drafted by GPT-4o, ensuring each entry reflects an authoritative recommendation rather than a model-generated one.

3.2 Human Verification of Zero-Shot Annotations

To assess the reliability of automated cause scoring, we manually annotated 700 Bengali samples

independently of the GPT-4o pipeline and computed inter-annotator agreement between human and model labels using Cohen's κ (Cohen, 1960), obtaining $\kappa = 0.85$, indicating substantial agreement. This validation step is a deliberate departure from prior work that treats LLM annotation as ground truth without independent verification (Gilardi et al., 2023); we treat GPT-4o as a scalable first-pass annotator whose output requires human auditing before being used as training supervision.

3.3 Cause Scoring with GPT-4o

We use GPT-4o (OpenAI, 2024) for zero-shot cause scoring across both languages. This choice is deliberate rather than default: at the time of annotation, GPT-4o offered the strongest documented multilingual and instruction-following performance among readily available API models for low-resource Indic-language text, where annotation quality depends heavily on the model's ability to interpret culturally specific cause categories (e.g., distinguishing brick-kiln pollution from general industrial emissions) without task-specific fine-tuning. We do not treat this as a claim of optimality across all possible models; rather, GPT-4o was selected under practical constraints of annotation cost, API stability for batch processing, and the need for consistent zero-shot behavior across two languages using an identical prompt template, and its labels are independently validated against human annotation (§3.2) rather than trusted outright.

3.4 Hindi Dataset Construction

No cause-annotated Hindi climate news resource exists at the time of writing (§2). We construct one by collecting 2,000 Hindi-language climate news articles from Down To Earth Hindi[1], balanced across the 16 cause categories established for Bengali (§3.1). Causes are scored using the same GPT-4o zero-shot pipeline and prompt template used for Bengali (§3.3), ensuring that any performance difference observed under cross-lingual transfer (§4.2) reflects the transfer setup itself rather than an inconsistency in annotation procedure between languages.

We use this dataset exclusively for zero-shot cross-lingual evaluation rather than fine-tuning: our best Bangla-trained classifier (§3.5) is evaluated directly on Hindi test data with no Hindi-language gradient updates. This design is intentional – the goal is to measure whether a taxonomy and classifier developed for one Indic language generalize to a related one without additional supervision, isolating the cross-lingual transfer question from the separate (and already well-studied) question of whether fine-tuning on Hindi data improves Hindi performance.

3.5 Benchmark Models

We benchmark five transformer backbones on the multi-label cause classification task, chosen to span distinct pretraining philosophies rather than to maximize raw model count:

- **BanglaBERT** (Bhattacharjee et al., 2022) – a monolingual encoder pretrained exclusively on Bengali text. Included as an upper-bound reference for language-specific pretraining; by construction it cannot support cross-lingual transfer to Hindi, making it a useful contrast against the multilingual backbones below.
- **MuRIL** (Khanuja et al., 2021) – pretrained specifically on Indian languages with explicit translation and transliteration augmentation between language pairs, designed to improve cross-lingual robustness across Indic scripts.
- **XLM-RoBERTa** (Conneau et al., 2020) – a general-purpose encoder pretrained across 100 languages without Indic-specific data curation. Included as a non-Indic-specialized multilingual baseline, to test whether broad multilingual coverage alone is sufficient for this task.
- **shihab17 Bangla-SBERT** (Uddin et al., 2024) – an XLM-RoBERTa backbone further finetuned via multilingual knowledge distillation from stsb-xlm-r-multilingual for Bengali sentence embedding (HuggingFace: shihab17/bangla-sentence-transformer). Included to test whether a sentence-similarity-oriented multilingual objective, rather than masked-language-model pretraining alone, transfers usefully to multi-label cause classification.
- **IndicBERTv2** (Doddapaneni et al., 2023) – pretrained on IndicCorp v2 across 24 Indic languages including both Bengali and Hindi. Included as the Indic-specialized multilingual

[1] https://hindi.downtoearth.org.in/

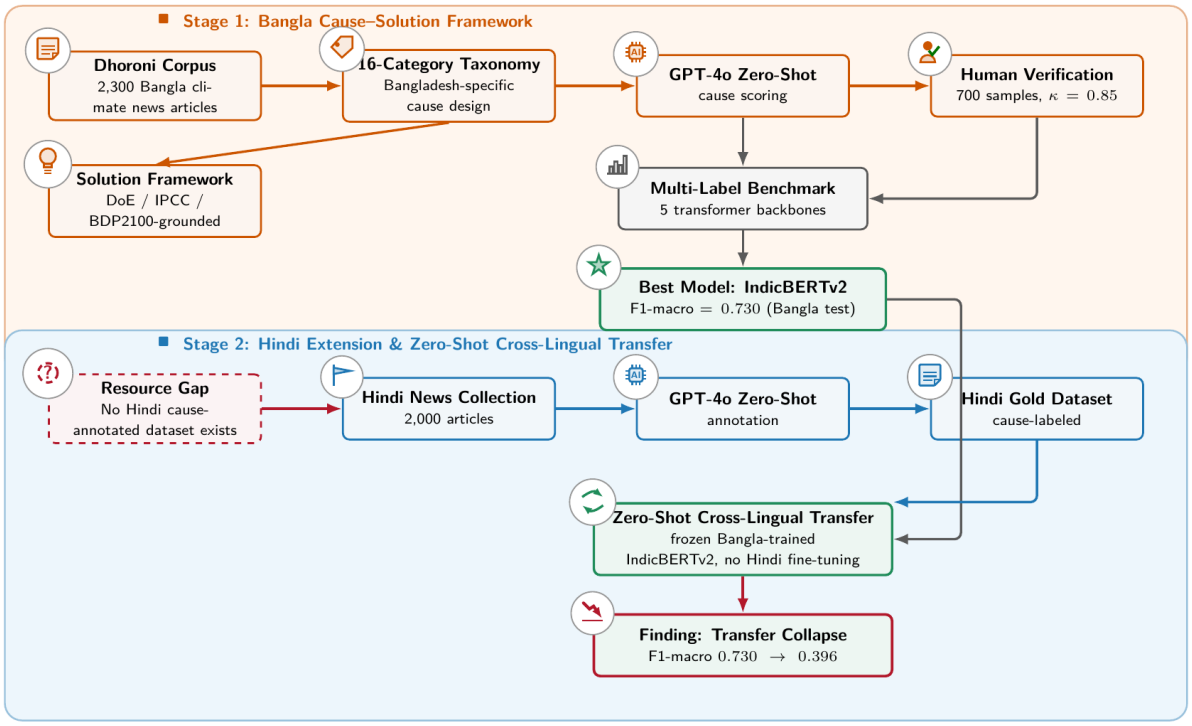


Figure 1: Overview of our two-stage methodology. **Stage 1 (top):** we extend the Dhoroni corpus (Wasi et al., 2024) with a 16-category, Bangladesh-specific cause taxonomy; causes are scored zero-shot with GPT-4o (OpenAI, 2024) and validated against 700 human-annotated samples (Cohen's $\kappa = 0.85$; Cohen, 1960). Each cause is paired with a source-grounded solution (DoE / IPCC / BDP2100). Five transformer backbones are benchmarked on the resulting multi-label task, with IndicBERTv2 (Doddapaneni et al., 2023) performing best (F1-macro $= 0.730$). **Stage 2 (bottom):** motivated by the absence of a comparable Hindi resource, we construct a 2,000-article Hindi climate news dataset annotated via the same GPT-4o pipeline, and evaluate the frozen, Bangla-trained IndicBERTv2 on it zero-shot, with no Hindi fine-tuning. F1-macro drops from 0.730 to 0.396 under transfer, our central empirical finding.

model most directly relevant to our cross-lingual transfer question, and the backbone we ultimately select for the zero-shot Hindi evaluation.

All five models are fine-tuned as multi-label classifiers on the Bengali training split with per-class threshold tuning on the validation set (§3.5), and evaluated on a held-out Bengali test set (Table 1). The best-performing model by F1-macro is subsequently frozen and evaluated zero-shot on the Hindi test set (§3.4).

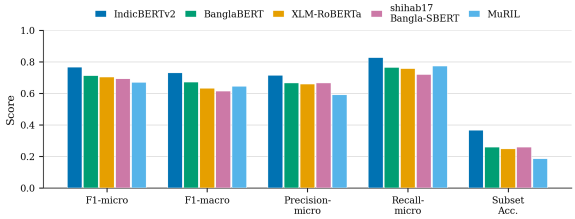


Figure 2: Per-metric comparison of all five backbones on the Bangla test set.

4 Results

4.1 In-Language Bangla Performance

IndicBERTv2 is the strongest backbone on every metric (Table 1; F1-macro 0.730, F1-micro 0.766). BanglaBERT is the best monolingual alternative (F1-macro 0.671) and beats three multilingual baselines despite no exposure beyond Bengali, suggesting monolingual pretraining depth substantially closes the gap with Indic-specialized multilingual pretraining in-language. MuRIL trails on precision (0.591) despite competitive recall (0.773) – the weakest precision/recall balance of the five. As Figure 2 shows, the five models cluster tightly on recall (0.72–0.83) but diverge sharply on subset accuracy (0.19–0.37): they largely agree on *which* causes are present, and differ chiefly in identifying the *full* co-occurring set per article.

Model	F1-micro	F1-macro	Precision-micro	Recall-micro	Subset Accuracy
IndicBERTv2	**0.766**	**0.730**	**0.713**	**0.827**	**0.366**
BanglaBERT	0.712	0.671	0.666	0.764	0.259
XLM-RoBERTa	0.704	0.632	0.659	0.756	0.247
shihab17 Bangla-SBERT	0.692	0.614	0.666	0.720	0.259
MuRIL	0.669	0.644	0.591	0.773	0.186

Table 1: Bangla test-set performance (per-class tuned thresholds). **Bold** = best result.

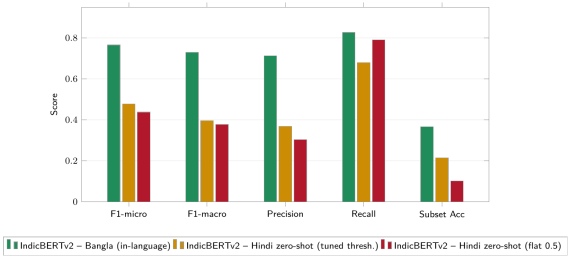


Figure 3: Recall is preserved under transfer; precision and subset accuracy collapse – the signature of over-triggering, not lost cause-signal recognition.

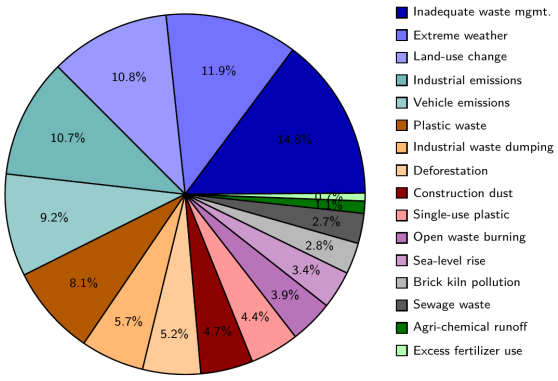


Figure 4: Distribution of the 16 cause categories across the Bangla corpus (2,300 articles, multi-label).

4.2 Why Zero-Shot Transfer to Hindi Fails

We freeze IndicBERTv2 and evaluate it on the 2,000-article Hindi gold dataset with no Hindi gradient updates (Table 2). F1-macro drops from 0.730 to 0.396 – a 46% relative decrease. The cause is not what a "cross-lingual transfer fails" headline usually implies. **Recall is preserved**: 0.827 (Bangla) $\rightarrow$ 0.679 (Hindi, tuned) $\rightarrow$ 0.791 (Hindi, flat 0.5). **Precision collapses**: 0.713 $\rightarrow$ 0.368 $\rightarrow$ 0.303. The model still recognizes when Hindi text expresses a cause category – it has not lost the underlying signal. What breaks is calibration: decision boundaries tuned on Bangla validation data do not hold on Hindi, and the model over-triggers categories at a rate that erodes precision far faster than recall degrades. Figure 3 makes this asymmetry visible directly.

4.3 Per-Category Findings

Transfer is not uniform, and the pattern is informative. *Brick-kiln pollution* collapses completely (Bangla F1 $= 0.82 \rightarrow$ Hindi F1 $= 0.00$), the single largest drop – consistent with it being one of the rarest Bangla categories (2.8% of instances; Figure 4), so the source model was already weakly calibrated on it before transfer was attempted. *Vehicle emissions* shows the classic over-triggering pattern: recall stays high (0.88) while precision falls to 0.16.

One category runs *opposite* to the general trend: *sewage waste* improves under transfer (Bangla F1 $= 0.41 \rightarrow$ Hindi F1 $= 0.63$), likely reflecting its far larger Hindi test support (132 vs. 17 in Bangla) rather than better cross-lingual representation quality.

The most specific finding is a **label confusion between two semantically adjacent categories**. In Bangla, *excess fertilizer use* has high precision (1.00) and *agricultural chemical runoff* has moderate precision (0.60). Under Hindi transfer, this **inverts**: agricultural chemical runoff jumps to precision 0.87 (recall collapses to 0.19), while excess fertilizer use falls to precision 0.15 (recall rises to 0.70). The two categories effectively swap their precision/recall profile. This is not random degradation – it is consistent with the model conflating two fine-grained, semantically overlapping classes once Bangla-specific lexical cues that separated them no longer transfer, defaulting toward one category at the other's expense. It suggests that future cross-lingual cause taxonomies may need coarser category boundaries, or explicit negative examples separating near-duplicate classes, to survive zero-shot transfer.

Setting	F1-micro	F1-macro	Precision-micro	Recall-micro
Bangla (in-language)	**0.766**	**0.730**	**0.713**	0.827
Hindi zero-shot, tuned thresh.	0.478	0.396	0.368	0.679
Hindi zero-shot, flat 0.5	0.438	0.377	0.303	**0.791**

Table 2: IndicBERTv2: Bangla in-language vs. zero-shot Hindi transfer (highlighted).

Source	Instances	Share
DoE	6,477	35.5%
Local guidance	3,825	21.0%
CIRC	3,286	18.0%
BDP2100	3,102	17.0%
IPCC	1,533	8.4%

Table 3: Source distribution across all 18,223 solution instances in the Bangla cause-solution framework.

4.4 Solution Framework Coverage

Beyond classification performance, the cause-solution framework itself (§3.1) is an empirical contribution we can characterize directly. All 16 cause categories have at least one grounded solution attached, with an average of 3.1 distinct solutions per category (range: 3–4). 96.7% of articles with at least one active cause label (2,223 / 2,300) have at least one corresponding solution attached, and articles average 8.0 solution instances each, reflecting that a typical article names multiple co-occurring causes, each with multiple grounded solutions.

Table 3 reports the source distribution across all 18,223 solution instances. DoE guidance is the largest single source (35.5%), followed by locally-sourced Bangladesh-specific guidance (21.0%), circular-economy guidance covering waste segregation and waste-to-energy conversion (CIRC; 18.0%), BDP2100 (17.0%), and IPCC (8.4%). The framework is not dominated by a single source type, and no cause category relies on only one source, supporting the traceability goal described in §3.1.

5 Conclusion

We introduced a 16-category, Bangladesh-specific cause taxonomy for Bengali climate news, extending the Dhoroni corpus, and paired each cause with a source-grounded solution – to our knowledge, the first cause-and-solution annotation framework for Bengali climate discourse. Causes are scored via GPT-4o zero-shot annotation and validated against 700 human annotations (Cohen's $\kappa = 0.85$). Across five transformer backbones, IndicBERTv2 performs best (F1-macro 0.730), and monolingual BanglaBERT closes much of the gap with multilingual alternatives in-language.

Motivated by the absence of a comparable Hindi resource, we constructed and release the first Hindi climate cause dataset of its kind (2,000 articles) and used it to test zero-shot cross-lingual transfer of our best Bangla-trained classifier. Transfer largely fails (F1-macro 0.730 $\rightarrow$ 0.396), but not for the reason a simple headline number suggests: recall is preserved while precision collapses, indicating a calibration failure rather than a loss of underlying cross-lingual signal, and we identify a specific case of label confusion between semantically adjacent categories that offers a concrete direction for more transfer-robust taxonomy design.

We hope this framework, and the negative transfer result in particular, are useful both as a resource for Bengali and Hindi climate NLP and as a cautionary data point against assuming that multilingual pretraining alone guarantees cross-lingual generalization on structured, culturally grounded classification tasks.

Limitations

- **Hindi labels rely on GPT-4o annotation without a matched human-verification round.** Bangla annotations are validated against 700 human-labeled samples (Cohen's $\kappa = 0.85$); the Hindi gold set has no equivalent audit. We note, however, that the transfer-collapse pattern we report is precision-specific with recall largely preserved (§4.2) – a signature consistent with threshold miscalibration under transfer, not with the more symmetric precision/recall degradation that would be expected from random annotation noise. Independently verifying a Hindi human-annotated subset is a natural next step to confirm this directly.

- **We isolate the zero-shot transfer setting by design.** We do not fine-tune on Hindi data, since our goal is to measure whether a Bangla-trained classifier generalizes without additional supervision. This means our results characterize zero-shot transfer specifically, not the ceiling achievable with Hindi fine-tuning – a complementary question we leave to future work.

- **The taxonomy is intentionally Bangladesh-specific.** It is designed around Bangladesh's environmental exposure profile rather than as a general-purpose South Asian taxonomy, and would need adaptation, not direct reuse, for other regional contexts.

- **Annotation and evaluation build on a single LLM (GPT-4o) and a single transfer language pair (Bangla→Hindi).** Both choices keep the study focused and internally consistent, but mean our findings on annotation reliability and cross-lingual transfer are demonstrated for this setup rather than established as fully general claims across other LLMs or Indic language pairs.

- **Solutions are source-grounded but not separately quality-rated.** Each solution is tagged with its originating source (DoE / IPCC / BDP2100 / local / circular-economy guidelines) for traceability, which is distinct from a human judgment of practical specificity – a natural extension for future annotation rounds.

Reproducibility. We will publicly release our code, the Bangla and Hindi cause-solution datasets, and all model checkpoints upon publication, to support reproduction and extension of this work by future low-resource South Asian climate NLP research.

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